

Assessment & Grading

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How Grading Works — Specifications Grading

Every assignment is graded pass / not-yet, never with partial points. Each one comes with a short, explicit list of standards (“specifications”) you can see before you start. If your work meets all of them, it passes. If it doesn’t yet, you get specific feedback and **revise and resubmit** — “not-yet” is a step in the process, not a penalty. You are never averaged down for an early attempt. Within each component the path to the grade you want is simply to meet the standards on more of the work — no partial credit, no curving inside an assignment. The components combine into the course grade by the weights in the [Syllabus](#) (§3.1.3): **Projects 55% · Homework 25% · Peer-engagement loops & discussions 10% · Checkpoints 9% · Prerequisite quiz 1%.**

This page details the **Homework component** (its deliverables, bundles, revise budget, and ladder) and summarizes the other components.

Homework: The 7 Deliverables, Best 5 Count

The homework component (25%) is **7 deliverables**, each pass / not-yet, with a built-in **drop**: only your best five core units count, so one core unit can be skipped or missed without touching your homework letter.

- **U0 — Tooling & Reproducibility** — the onboarding **completion floor**. It doesn’t by itself raise your letter, but every grade above E requires it.
- **Six core units** — **U1, U2a, U2b, U3, U4, U5** — the rungs of the ladder; the best five count.

Unit-by-unit descriptions, objectives, and rubrics are posted in **D2L** (the units are summarized on the [Homeworks](#) page). Assignment materials — starter code, autograder, data — are distributed via **Classroom 50**; submissions are pushed to your private repo, visible only to you and the instructor.

What it takes to pass a unit

A unit passes only when **all three bundles** are met:

1. **Correctness** — autograded method-validity / outcome specs (the core gateway).
2. **Interpretation (CERL)** — Claim, Evidence, Reasoning, Limits — human-confirmed.
3. **Process** — reproducibility and the submission contract.

Miss a bundle and the unit is **not-yet**: you revise *that bundle* and resubmit, not the whole unit.

The Homework Ladder

The homework component's grade is set by a straight **A B C D E** ladder (no plus/minus) — the per-component ladder the [Syllabus](#) §3.1.3.2 describes. **U0 must be passed in every case above E.**

Letter	Requirement
A	any 5 core units
B	4 core units
C	3 core units
D	2 core units
E	1 or fewer core units, or U0 not passed

The drop is automatic: with six core units and five rungs to the A, the lowest (or skipped) unit simply does not count. There is nothing to declare.

Revise & Resubmit Budget

- **Correctness (autograded): unlimited resubmissions** until the unit's deadline — the autograder re-runs on every push at no human cost, so keep pushing until it passes.
- **Interpretation + Process (human-confirmed): one** revise-and-resubmit per unit, **plus a course-wide pool of 3 tokens** for a second human-bundle revision on the units you find hardest.

Dates: week pacing; this term runs 7.5 weeks with no fixed calendar dates. Deadlines and post-feedback revision windows are posted per offering on D2L.

Other Course Components

Weights per the [Syllabus](#) §3.1.3: Projects 55%, Peer-engagement loops & discussions 10%, Checkpoints 9%, Prerequisite quiz 1%. Each is specs-graded internally (pass / not-yet with revision where the format allows).

Projects

One project replaces traditional midterm and final exams: two parts on one clinical dataset, nine milestones, gated by the checkpoint quizzes. Part 1 is due **Week 5 (the midpoint)**; Part 2 closes at **Week 7.5**.

Part 1: “One Problem, Three Lenses” — *Assigned Week 1 · due Week 5, the midpoint*

Use a single regression problem with a clinical dataset. Solve it three ways:

- **Least Squares:** classical normal equations and interpretation
- **Maximum Likelihood Estimation:** Gaussian model, negative log-likelihood
- **Bayesian:** posterior inference and credible intervals

Synthesis question: How do the three approaches diverge and converge? What assumptions differ?

Part 2: “From Inference to Discovery” — *Begins at the midpoint · due Week 7.5*

Same dataset, now tasked with discovering structure:

- **Approximate Inference:** Bayesian logistic regression — Newton/MAP, the Laplace approximation, and a Metropolis sampler
- **Discriminative Classification:** soft-margin SVM vs. the probabilistic classifier
- **Unsupervised Learning:** k-means clustering + PCA for dimensionality reduction

Synthesis question: What story does the data tell when you stop presuming its structure?

How a project is judged

Each milestone lists its **Satisfactory criteria at the top of its own notebook**. All criteria must be met: it is all-or-nothing, never partial credit. A **Not-Yet** returns with targeted feedback and **one revise cycle** back to Satisfactory. A project is complete when all its milestones are Satisfactory and all its gating checkpoints are passed.

Part 2 is cumulative but not weighted more heavily than Part 1, because specs grading already requires that Part 2 mastery presupposes Part 1's foundations.

Bundles and gates

Part 1 is Satisfactory when 1.1, 1.2, 1.3, and 1.4 are all Satisfactory and Checkpoint quizzes 1.1, 1.2, and 1.3 are passed. The structured project is complete when both parts are.

Part 2 is organized as two thematic gates plus the synthesis:

Bundle	Requires	Gated by
Gate A — Supervised	2.1 (approximate inference) and 2.2 (classification) Satisfactory	Checkpoint quiz 2.1
Gate B — Unsupervised	2.3 (clustering) and 2.4 (dimensionality reduction) Satisfactory	Checkpoint quiz 2.3
Synthesis	2.5 Satisfactory	structured peer review

There is no separate checkpoint for 2.2 or 2.4: each gate carries one checkpoint covering the pair.

Submission: Classroom 50 (the invite link is posted on D2L at term start).

Checkpoints

Five short open-book **D2L quizzes** (about 20 to 30 minutes each) on each module's derivations, and they **gate the project milestones**: each quiz is due the week its module is taught, and the milestone it gates is due the following week. Most items auto-grade on submission; each quiz also carries two or three short written responses graded Satisfactory / Not-Yet with one revision.

The coding verification that used to sit beside these derivations moved to where the code already lives. Homework Units 1, 3, and 4 carry the checkpoint 1.1, 1.2, and 1.3 verification exercises; the 2.1 and 2.3 verifications were already required inside milestones 2.1, 2.3, and 2.4. Nothing is assessed twice.

Quiz due	Checkpoint	Gates
Week 2	1.1 — Normal equations	Milestone 1.1 (Wk 3)
Week 3	1.2 — MLE for the weights	Milestone 1.2 (Wk 4)
Week 4	1.3 — Identify the conjugate posterior	Milestone 1.3 (Wk 5)

Quiz due	Checkpoint	Gates
Week 5	2.1 — Newton–Raphson update	Milestone 2.1 (Wk 6) · Gate A
Week 6	2.3 — k-means objective & PCA	Milestone 2.3 (Wk 7) · Gate B

Week pacing; this term runs 7.5 weeks with no fixed calendar dates.

Peer-Engagement Loops

A loop runs every week. Week 3 has no tool, so its written discussion carries the loop alone; Week 5 has no discussion (it is the project midpoint), so its loop is the tool and mastery check alone. Each full loop: complete the interactive tool and pass its built-in mastery check (the loop’s knowledge check), post your reflection + token to your team’s D2L discussion, give peer review to **every member of your team**, then respond to the reviews on your post and revise. **Credit requires all stages, including replies to all teammates.** See [Activities](#) for loop details.

Prerequisite Quiz

A brief diagnostic (Week 1, 20 minutes, closed book, 33 points) on the three course prerequisites: **univariate calculus** (derivatives of composed functions, definite integrals), **linear algebra** (transposes, matrix dimensions and validity of products, inner products), and **basic Python and command line** (a short function, a few shell invocations). You write your answers out and show your work on the math, typed into the answer box or attached as a photo or scan of handwritten work, so credit is awarded rule by rule rather than on the final answer alone.

It exists to surface gaps early, not to gatekeep: most students pass, and the score is worth 1%. A description document posted on D2L beside the quiz lists every skill tested, section by section, with review resources for each, so a low score comes with a concrete plan.